

Element H: Prototype Testing and Data Collection Plan

The development office at our independent boarding and day school has an overflow of items. The excess items piled on the floor hinder workflow and efficiency. Many items move around the development office; therefore, it is important to keep them organized. The current storage situation is unsatisfactory in the eyes of school donors and visitors.

Our testing plan will address roughly 80% of the design criteria, including weight load capacity, storage space, and whether or not there is access to outlets, as well as a possible coffee station. The testing procedures are as follows:

Outlets

1. Measure the distance of the power outlet from the wall and floor.
2. Compare the current design and see if outlets will be accessible.
3. Display the height data of the socket on the modeling image. We can choose to hollow out the back of the board or not use the board to cover it.
4. If it is in a socket that is difficult to connect to. We have to use the extended power strip.

The wires of the power strip can be fixed under the desktop, which won't affect the appearance either. These are all the engineering projects that can be supplemented at the end.

Storage space

1. Measure boxes, storage bins, and all other items that will be stored.
2. Compare the current design and see if size needs to be adjusted.
3. Have a certain range of understanding of the number of boxes. This is helpful for the data changes of the design.

Concealed boxes

1. Measure items that will need to be concealed.
2. Compare to the current design. Try to assume whether a box can be hidden when placed in this space
3. Record data and adjust design accordingly.

Coffee station

1. Measure the minimum coffee station dimensions.
2. Check whether it is convenient to connect the socket.
3. Examine the area of the coffee station. The required area cannot be larger than the designed area. If the design data needs to be changed, it is necessary to discuss and implement it with others.

Measurements of the coffee station will ensure it fits and is positioned near a power source. The measurement of the outlet's location relative to the floor and far wall will allow us to compare its location to the planned location of the design and confirm there will be no overlap. The measurements of the boxes and storage bins allow us to calculate shelf volume and ensure sufficient storage. Weight capacity estimates from material suppliers will confirm the design is safe and sturdy. The remaining open space that has not been allocated to another requirement can be used as display shelving. The procedures we made will test our three highest priority requirements.

The expert we chose to evaluate our testing procedures was Ms. Z. She was chosen due to her extensive experience overseeing student design projects and her strong understanding of the design process. Her feedback ensures that our testing approach is thorough and realistic. She reviewed our testing plan and made important improvements. She added tests for high-priority requirements, like the concealed storage boxes, and included a step to record data during testing.

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She also reorganized the plan to make it clearer and easier to follow.